

Homework 2

March 9, 2026

Problem 1

Answer the following questions. For questions requiring experiments in C language, compile your code with the GCC compiler. The code and output should be included.

- (a) In a regular C program, what value does the expression `0.0` convey? `+0.0` or `-0.0`? Please find the statement in the manual

<https://www.gnu.org/software/gnu-c-manual/gnu-c-manual.html>

that supports your answer and perform experiments with a C program.

- (b) According to the C99 standard, what function (or macro) from the standard libraries can we use to get the sign bit of a floating-point number? Find statements from the C99 standard to support your answers.

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n1256.pdf>

- (c) How do we specifically assign `+0.0` and `-0.0` in C language? Experiment with C codes and validate the assigned value using the function you found in (b).
- (d) Suppose we have two floating-point numbers

$$a < 0 \text{ and } b, \text{ where } b \text{ is a number that is neither NaN nor } \pm \infty.$$

Also, we have a C program that contains the following line:

$$c = a / \max(b, 0.0);$$

We wish to guarantee that

$$c < 0$$

always holds (You can assume that b is not too large, so no underflow occurs when calculating c).

Which implementation should we use for the “max” function? Explain your choice and validate your answer with experiment.

(1)

$$(x > y) ? x : y$$

(2)

$$(x < y) ? y : x$$

Hint: “`max(b, 0.0)`” should not return any **negative** number.

Problem 2

Use Matlab for Cholesky Factorization

In this homework, you need to implement Cholesky factorization in outer product form using **MATLAB**. If you do not have MATLAB, use **Octave** instead. In the outer product form, your procedure recursively calculates a smaller matrix:

$$\begin{bmatrix} \alpha & \mathbf{v}^T \\ \mathbf{v} & \mathbf{B} \end{bmatrix} = \begin{bmatrix} \sqrt{\alpha} & \mathbf{0}^T \\ \frac{\mathbf{v}}{\sqrt{\alpha}} & \mathbf{I} \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{B} - \frac{\mathbf{v}\mathbf{v}^T}{\alpha} \end{bmatrix} \begin{bmatrix} \sqrt{\alpha} & \frac{\mathbf{v}^T}{\sqrt{\alpha}} \\ \mathbf{0} & \mathbf{I} \end{bmatrix} = \begin{bmatrix} \sqrt{\alpha} & \mathbf{0}^T \\ \frac{\mathbf{v}}{\sqrt{\alpha}} & \mathbf{L} \end{bmatrix} \begin{bmatrix} \sqrt{\alpha} & \frac{\mathbf{v}^T}{\sqrt{\alpha}} \\ \mathbf{0} & \mathbf{L}^T \end{bmatrix},$$

where

$$\mathbf{B} - \frac{\mathbf{v}\mathbf{v}^T}{\alpha} = \mathbf{L}\mathbf{L}^T \quad (1)$$

When updating the matrix

$$\mathbf{B} - \frac{\mathbf{v}\mathbf{v}^T}{\alpha},$$

consider these three implementations:

- No loop (vectorized): Use matrix/vector operations provided by Matlab to do

$$\mathbf{B} = \mathbf{B} - \frac{\mathbf{v}\mathbf{v}^T}{\alpha}.$$

- One-level loop: Update each column of $\mathbf{B}$ in a for loop. That is,

$$\mathbf{B}_{:,j} = \mathbf{B}_{:,j} - \mathbf{v} \frac{v_j}{\alpha}$$

for each j .

- Two-level loop: Update each element of $\mathbf{B}$ in two for loops. That is,

$$\mathbf{B}_{i,j} = \mathbf{B}_{i,j} - v_i \frac{v_j}{\alpha}$$

for each i and j .

Requirements

- Implement** the three above-mentioned versions and **include the code** for all experiments in your report.
- Generate some $n \times n$ symmetric **positive definite** matrix and then verify your program results with the built-in function “chol”. Explain how you generated the positive definite matrices.
- Compare the efficiency of your variants under different n , where we suggest you at least include $\{10, 100, 1000, 4000\}$ in the range of n , and measure the running time (elapsed time). Depending on the power of your computer, the range of n can be smaller or larger. For timing, you can check

```
>> help toc
```
- Explain the results of (iii). Hint: Read about “vectorization” for MATLAB/Octave.
- (**Extra** 10 pts) If you used MATLAB, do (iii) again in Octave and discuss whether the same thing happens. If you used Octave, do the same thing with MATLAB instead. The extra points are added to the total score of HW2, but the total score of HW2 will not exceed 100 pts.